



Department of Artificial Intelligence & Machine Learning

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Date: 06th August 2026.

EXPERIMENT NO. 4

AIM: To implement:

1. High Pass Filter, and
2. High Boost Filter.

THEORY:

1. Basic High Pass Filter

The principal objective of high pass (sharpening) filter is to highlight fine detail in an image or to enhance detail that has been blurred, either in error or as a natural effect of a particular method of image acquisition. Uses of image sharpening vary and include applications ranging from electronic printing and medical imaging to industrial inspection and autonomous target detection in military systems.

The shape of the impulse response needed to have a high pass (sharpening) spatial filter indicates that the filter should have positive coefficients in the outer periphery. For a 3 x 3 mask, choosing a positive value in the centre location with negative coefficients in the rest of the mask meets this condition. Thus when the mask is over an area of constant or slowly varying gray level, the output of the mask is zero or very small. This result is consistent with what is expected from the corresponding frequency domain filter.

-1	-1	-1
-1	8	-1
-1	-1	-1

Fig: A high Pass {Laplacian} filter mask

2. High Boost Filtering



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A process used for many years in the publishing industry to sharpen images consists of subtracting a blurred version of an image from the image itself.

This process, called unsharp masking, is expressed as:

$$f_s(x,y) = f(x,y) - f'(x,y)$$

where, $f_s(x,y)$ denotes the sharpened image obtained by unsharp masking and $f'(x,y)$ is a blurred version of $f(x,y)$. The origin of unsharp masking is in dark room photography, where it consists of clamping together a blurred negative to corresponding positive film and then developing this combination to produce a sharper image.

A further generalization of unsharp masking is called high-boost filtering. A high-boost filtered image, f_{hb} , is defined at any point (x, y) as

$$f_{hb}(x, y) = A f(x,y) - f'(x,y)$$

where $A \geq 1$ and as before $f'(x,y)$ is the blurred image of $f(x,y)$

The final expression for high-boost image is given as

$$f_{hb}(x, y) = (A-1) f(x,y) + f_s(x,y)$$

The above given equation is applicable in general and does not state explicitly how the sharp image is obtained.

The mask for high-boost filtering is given as:

-1	-1	-1
-1	A+8	-1
-1	-1	-1

Fig: A mask for High-Boost filtering

One of the principal applications of high-boost filtering is when the input image is darker than desired. By varying the boost co-efficient, it generally is possible to obtain an overall increase in average gray level of the image, thus helping to brighten the final result.



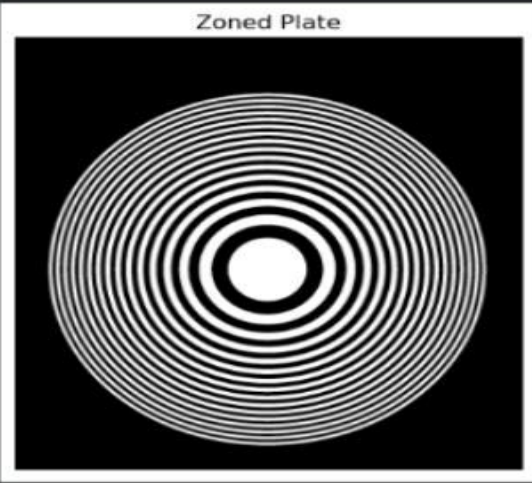
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Code & Output:

```
import cv2
from matplotlib import pyplot as plt
import numpy as np

img = cv2.imread(r"C:\Users\djsce.student\Desktop\zone_plate.jpg")

if img is not None:
    img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    plt.imshow(img)
    plt.title("Zoned Plate")
    plt.axis("off")
    plt.show()
else:
    print("Error")
```

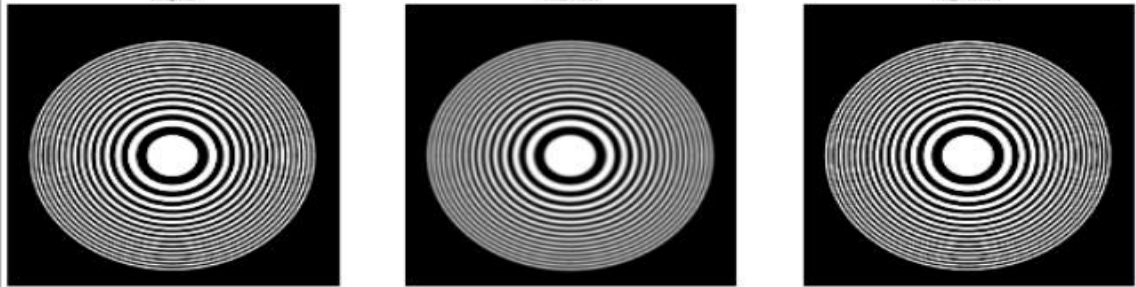


```
A = 1.5

low_pass = cv2.GaussianBlur(img, (5, 5), 0)

high_boost = cv2.addWeighted(img, A, low_pass, -(A-1), 0)
high_boost = np.clip(high_boost, 0, 255).astype(np.uint8)

plt.figure(figsize=(20, 10))
plt.subplot(1, 3, 1); plt.imshow(img, cmap='gray'); plt.title('Original'); plt.axis('off')
plt.subplot(1, 3, 2); plt.imshow(low_pass, cmap='gray'); plt.title('Low Pass'); plt.axis('off')
plt.subplot(1, 3, 3); plt.imshow(high_boost, cmap='gray'); plt.title('High Boost'); plt.axis('off')
plt.show()
```

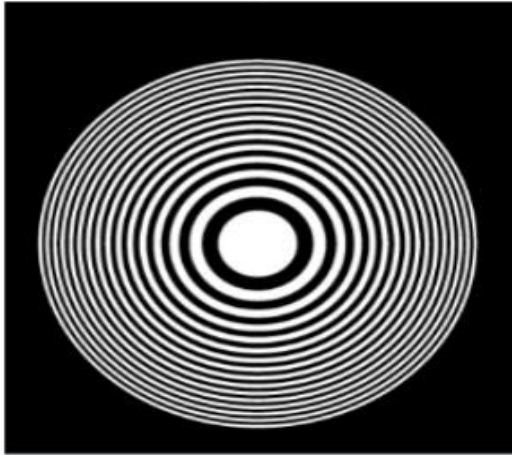


```
hp_kernel = np.array([[[-1, -1, -1], [-1, 8, -1], [-1, -1, -1]]])
high_pass = cv2.filter2D(img, -1, hp_kernel)

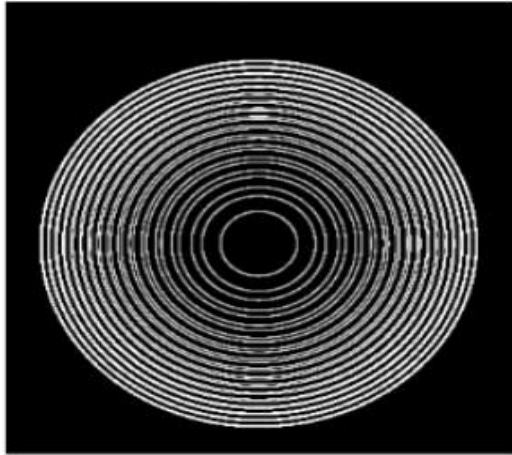
plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1); plt.imshow(img, cmap='gray'); plt.title('Original'); plt.axis('off')
plt.subplot(1, 2, 2); plt.imshow(high_pass, cmap='gray'); plt.title('High Pass'); plt.axis('off')
plt.show()
```

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Original



High Pass



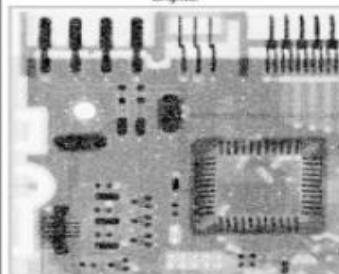
```

img = cv2.imread(r"C:\Users\djsce.student\Desktop\pcb.jfif")
low_pass = cv2.GaussianBlur(img, (5, 5), 0)

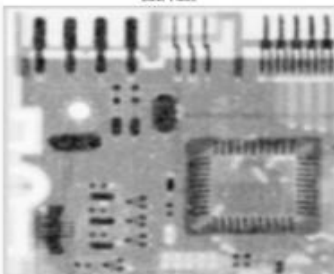
high_boost = cv2.addWeighted(img, A, low_pass, -(A-1), 0)
high_boost = np.clip(high_boost, 0, 255).astype(np.uint8)

plt.figure(figsize=(20, 10))
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plt.subplot(1, 3, 3); plt.imshow(high_boost, cmap='gray'); plt.title('High Boost'); plt.axis('off')
plt.show()
  
```

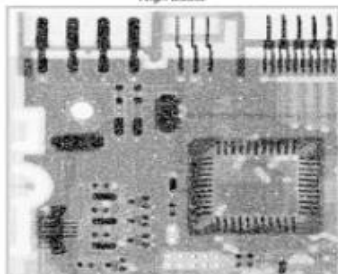
Original



Low Pass



High Boost

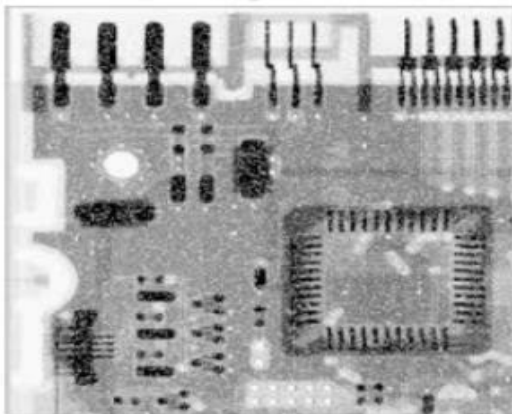


```

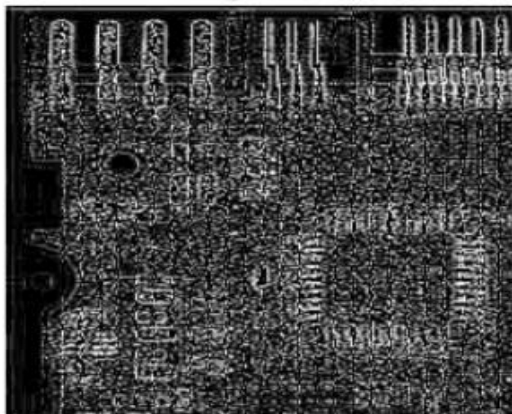
hp_kernel = np.array([[ -1, -1, -1], [-1, 8, -1], [-1, -1, -1]])
high_pass = cv2.filter2D(img, -1, hp_kernel)

plt.figure(figsize=(10, 5))
plt.subplot(1, 2, 1); plt.imshow(img, cmap='gray'); plt.title('Original'); plt.axis('off')
plt.subplot(1, 2, 2); plt.imshow(high_pass, cmap='gray'); plt.title('High Pass'); plt.axis('off')
plt.show()
  
```

Original



High Pass





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Post Lab Session:

Q. 1 Why do high-pass filters emphasize edges and fine details in an image?

Q2. What is the purpose of high-boost filtering, and how does it differ from simple high-pass filtering?

Q.3 Discuss the advantages and limitations of high-pass and high-boost filtering in real-world image enhancement applications.



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→ High-pass filters emphasize edges & fine details because they remove low-freq. components (smooth or uniform regions) & retain high-freq. components, where pixel intensity changes rapidly. Edges, lines, & textures correspond to these high-freq. changes.

→ The purpose of high-pass filtering is to sharpen an image while preserving its original appearance. It enhances fine detail by adding an amplified version of the high-freq. components back to the original image.

- High pass filter removes low-freq. information & mainly displays edges & fine details.
- High-pass filtering keeps the original image & strengthens edges.

→ High pass filters:

Advantages:

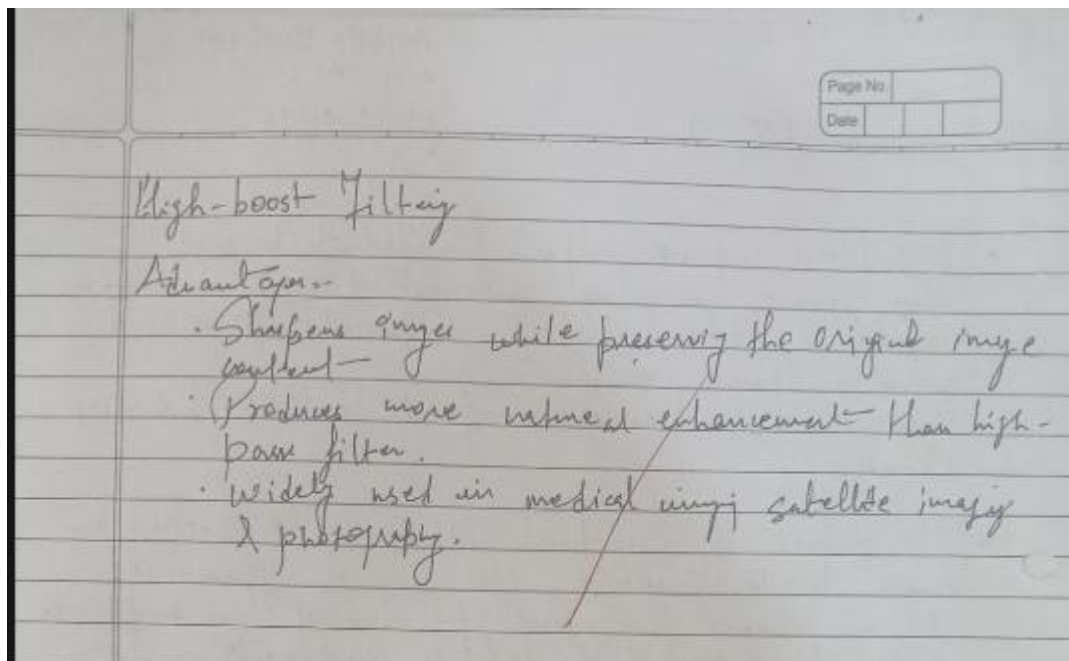
- Enhances edges & fine details.
- Useful for edge detection & feature extraction.
- Improves image sharpness.

Limitations:

- Amplifies image noise.
- Removes smooth regions & may lose important information.
- Can produce unnatural-looking images if overused.



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Conclusion:

The High Pass and High Boost filtering techniques were successfully implemented to enhance the fine details and edges of the image. The High Pass filter emphasized high-frequency components, making edges and sharp features more prominent. High Boost filtering further enhanced these details while also retaining the original image information. Thus, both techniques are effective for image sharpening and improving the visual quality of images.

REFERENCE: Digital Image Processing by Gonzalez and Woods, 2nd edition.